

# Stock Market Reactions to Supply Chain Disruptions and Recovery from the 2022 Shanghai Lockdown\*

Kyunghee Song

KAIST College of Business, Korea Advanced Institute of Science and Technology, South Korea

Yun Shin Lee\*\* 

KAIST College of Business, Korea Advanced Institute of Science and Technology, South Korea

Moon Su Koo

Memory Business, Memory Marketing Team, Samsung Electronics Co., Ltd., South Korea

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## Abstract

This paper investigates the effects of the 2022 Shanghai lockdown and its subsequent lifting on global supply chains during the COVID-19 pandemic. We first identify four focal firms—Tesla, Volkswagen, SAIC Motors, and General Motors—most frequently cited in the news articles as experiencing supply chain disruptions due to the lockdown. Using an event study methodology, we estimate abnormal stock returns for these focal firms and their tier-1 global suppliers following the lockdown and its lifting announcements. We find significant and systematic stock market reactions for the suppliers of the focal firms, whereas the focal firms themselves do not exhibit statistically significant stock reactions. Specifically, suppliers experienced significantly negative abnormal returns during the lockdown, with the opposite effect observed upon the lifting of the lockdown. Furthermore, supplier characteristics such as capital expenditure and inventory turnover significantly influenced cumulative abnormal returns during the lockdown, while leverage and R&D intensity affected returns during the lifting event. Compared to the suppliers of benchmark firms, those of the focal firms experienced significantly more negative cumulative abnormal returns in the lockdown case, while no significant difference was observed between them in the lifting case.

**Keywords** Event study; Abnormal returns; COVID-19; Supply chain disruption; Supply chain recovery

*JEL Classification:* E44, G14

## 1. Introduction

The World Health Organization (WHO) first reported the outbreak of an unprecedented coronavirus in China at the end of 2019, later officially named COVID-19. Due to its highly infectious nature, the virus quickly spread across continents,

resulting in a global pandemic. In response, governments worldwide implemented various measures to control the spread of the virus, such as quarantine protocols and the isolation of individuals who tested positive (Yang *et al.* 2020). As a result, manufacturers across multiple industries faced significant challenges in sourcing materials and parts (Linton and Vakil 2020). According to the World Trade Organization (WTO), the value of global merchandise exports declined by 8% in 2020, indicating severe disruptions in global supply chains (World Trade Organization 2021).

After the initial surge of infection cases in China, the Chinese government implemented stringent countermeasures to control the spread of the virus, including the Wuhan lockdown in January 2020. These efforts successfully curbed the pandemic for a while. However, with the emergence of the Omicron variant, infection cases in China suddenly surged again in early March 2022 (Bradsher 2022). The Omicron variant was more transmissible, leading to a sharp increase in infection cases across China again (Bradsher 2022). As the variant spread to major cities, such as Changchun and Shenzhen, Chinese authorities responded by imposing strict restrictions on these cities (Jie and Kubota 2022; Kubota 2022).

On March 11, 2022, Shanghai authorities ordered the closure of theaters and museums and switched schools to online learning (Bradsher 2022). Despite these measures, the number of confirmed cases continued to rise. Initially, Shanghai officials resisted implementing a citywide lockdown due to its potential impact on national economic and social development (Jie *et al.* 2022; Mitchell *et al.* 2022). However, they repeatedly reversed the decision by announcing temporary, staged lockdowns. On April 4, 2022, an indefinite lockdown was unexpectedly announced (Fan 2022). A few months later, on May 15, 2022, Shanghai officials outlined plans to ease restrictions (Kahn 2022). However, they revised their statement the following day, announcing that the reopening process would begin on June 1, 2022 (Hua 2022).

This paper focuses on the lockdown imposed on Shanghai, a key hub for manufacturing and trade in China. Shanghai is home to significant operations of international automakers such as Tesla and Volkswagen, which maintain their supply chains through local plants. According to estimates by Bank of America, Shanghai accounted for 3.8% of China's GDP and was responsible for 7.2% of the country's total exports in 2021 (Gryta 2022). Although the pandemic, which first started in Wuhan, seemed to be under control due to the measures enforced by Chinese authorities, the Shanghai lockdown was subsequently implemented in response to the resurgence of infection cases due to the highly infectious variant. This lockdown provides a valuable case for examining how companies are influenced by the restrictions imposed by Shanghai authorities, particularly after their initial experiences in Wuhan. In May 2022, the *Financial Times* reported, "authorities raced to contain an outbreak of the highly infectious coronavirus variant Omicron. The repercussions have rippled across the globe, with multinationals from Apple, Tesla, and General Electric to Amazon, Adidas, and Estee Lauder warning of

disruption to their supply chains due to the lockdown of a city that handles 20 percent of China's international trade (Hollinger *et al.* 2022)". Given Shanghai's pivotal role in international manufacturing and the severe impact of the Omicron variant, this study focuses on the supply chain disruptions caused by the Shanghai lockdown.

We reviewed online news articles from three prominent sources, *The Wall Street Journal*, *Financial Times*, and *Reuters*, regarding the Shanghai lockdown. From this, we identified four focal firms—Tesla, Volkswagen, SAIC Motors, and General Motors—most frequently cited as experiencing supply chain disruptions due to the Shanghai lockdown. While our search was not limited by industry, all the selected focal firms are international automotive manufacturers with large factories in Shanghai, which serves as a key hub for their operations. For example, Tesla, the US-based electric vehicle manufacturer, produced more than half of its vehicles from its Shanghai Gigafactory in 2021 (Elliott 2022). We also expect that the Shanghai lockdown significantly impacted not only the focal firms but also their suppliers. Since global supply chains are interconnected, disruptions in one location can reverberate across the entire supply chain. For example, the Great East Japan Earthquake in 2011 negatively affected not only companies operating business in Japan but also companies located outside of Japan (Hendricks *et al.* 2020). Moreover, Jacobs and Singhal (2020) and Jacobs *et al.* (2022) show that an adverse event for an individual firm can hurt its global suppliers or customers. Therefore, we hypothesize that the Shanghai lockdown and its subsequent lifting affected not only focal firms in Shanghai but also their global supply chain partners. Jacobs and Singhal (2020) found that the propagation effect of the Volkswagen scandal on tier-2 suppliers was significantly weaker than on tier-1 suppliers. This suggests that the propagation effect weakens as it moves along the supply chain, leading us to focus on tier-1 suppliers in our analysis.

We use an event study methodology to measure the effects of the Shanghai lockdown on the financial performances of the focal firms and their first-tier suppliers. Specifically, we estimate the abnormal changes in stock prices surrounding the two critical events—the lockdown announcement and the announcement of its lifting. Our findings indicate that supply chains exhibited systematic reactions to both the disruption caused by the lockdown and the recovery following its lifting. Suppliers of focal firms experienced significant negative abnormal returns following the lockdown announcement, while they saw significant positive abnormal returns when the lockdown was lifted.

Moreover, we find that the impact of the lockdown on stock price changes varied based on the supplier's operational and financial characteristics. Suppliers that had made substantial capital investments in the year prior to the Shanghai lockdown experienced more negative stock price reactions. Additionally, suppliers with high inventory turnover rates were particularly vulnerable to the disruptions caused by the lockdown. Following the announcement of the lockdown's lifting, leverage

had a significantly positive effect on stock returns, while suppliers with high R&D expenditure over their sales received negative stock market reactions.

We also compare the performance of our focal firms and their suppliers with a set of benchmark firms—international automotive manufacturers of similar size that do not operate plants in Shanghai. The selected benchmark firms include Toyota, Stellantis, Ford Motor Company, Bayerische Motoren Werke AG (BMW), and Hyundai Motor Company. After excluding suppliers shared by both the focal firms and the benchmark firms, we find that the suppliers of focal firms were more negatively affected by the lockdown than those of the benchmark firms. However, the difference in abnormal returns between the two groups in response to the lockdown lifting was not statistically significant.

Several studies have used event study methodologies to investigate the effect of supply chain disruptions on firms' financial performance. These disruptions stem from various sources, including environmental issues (Jacobs and Singhal 2020), poor operations practices (Hendricks and Singhal 1997, 2003, 2009), and external disruptive events such as earthquakes and trade measures (Hendricks *et al.* 2020; Jacobs *et al.* 2022). Regarding the COVID-19 pandemic, earlier studies examined its initial impact on financial markets. He *et al.* (2020) and Ding *et al.* (2022) analyzed the effect of the Wuhan lockdown on Chinese stock market reactions using event study methodologies, while Pandey and Kumari (2021), Xie *et al.* (2022), Alam *et al.* (2021), and Srinivasan *et al.* (2023) focused on the impact of early pandemic announcements on global stock markets, including those in Australia and the United States. Furthermore, Xia *et al.* (2023), Wang *et al.* (2022), and Choudhury *et al.* (2022) investigated the propagation effects of supply chain disruptions during the early pandemic period. Unlike prior studies that focused on the early stages of the pandemic, this study examines the impact of the Shanghai lockdown and its subsequent lifting on global supply chains by analyzing the abnormal returns of global tier-1 suppliers of international automobile companies. Since this period represents a later phase of the pandemic, transitioning into a recovery stage after the initial infections were mitigated (Park *et al.* 2024), this study provides valuable insights into the long-term implications of pandemic-related supply chain disruptions.

The remaining sections of the paper are organized as follows. Section 2 reviews the relevant literature. Section 3 outlines the methodology for the empirical analysis. Section 4 describes the sample selection procedure, and Section 5 presents the results. Section 6 summarizes our findings and concludes.

## 2. Literature Review

### 2.1. Supply Chain Disruptions

A stream of literature uses event study methodologies to investigate the impact of supply chain disruptions on stock market reactions. Several studies examine how social and environmental events influence firms' financial performances (Jacobs

*et al.* 2010; Flammer 2013; Arora *et al.* 2020; Jacobs and Singhal 2020). Related to supply chains, for example, Jacobs and Singhal (2020) investigate the effect of the Volkswagen diesel emissions scandal on the global motor vehicle ecosystem, including Volkswagen's suppliers, customers, and competitors. They show that Volkswagen's tier-1 direct material suppliers experienced a 2.69% decline in average stock prices, while European customers of Volkswagen suffered an average drop of 5.28%. These findings suggest that the issues of a single firm have far-reaching consequences, affecting other companies in its supply chain.

Hendricks and Singhal (1997, 2003, 2009) examine the impact of firms' poor operations management on their financial performance. They also identify firm-specific and industry-specific factors that can influence the direction and magnitude of changes in shareholder value following negative operations management events. Hendricks and Singhal (1997) find that delays in introducing new products to the market result in an average market value decline of 5.25%. Hendricks and Singhal (2003) investigate the relationship between stock prices and supply chain glitches caused by inaccurate forecasts, shortages of parts, and poor planning. They show that supply chain glitch announcements lead to an average firm value loss of 10.28% for 2 days. Hendricks and Singhal (2009) further reveal that demand–supply mismatches due to too much supply compared to demand result in the average stock price change ranging from  $-6.79\%$  to  $-6.93\%$  over a 2-day window.

While earlier studies focus on disruptive events that occur internally within a firm, more recent research has shifted toward examining the effects of supply chain disruptions caused by external factors beyond a firm's control. Hendricks *et al.* (2020) investigate the financial impact of the supply chain disruptions caused by the 2011 Great East Japan Earthquake. They show both positive and negative financial results of the earthquake. Companies directly affected by the disruption experienced an average decline of 5.21% in stock prices within the first month, while competitors and firms in the rebuilding industry benefited, seeing an average increase of 4.79% in shareholder value. Jacobs *et al.* (2022) examine the supply chain disruptions driven by a policy change, specifically the US trade action against ZTE. They find that both ZTE's tier-1 and tier-2 US suppliers experienced a sharp decrease in shareholder wealth, while the opposite held for its competitors.

## 2.2. Prior Pandemic Event Studies

Several studies have examined the impact of supply chain disruption caused by the COVID-19 outbreak in China. He *et al.* (2020) estimate the effect of the Wuhan lockdown on the Chinese stock market on January 23, 2020, and compare how the effect varied across different industry sectors. They find that sectors such as transportation, mining, electricity and heating, and environment services suffered the most, while the manufacturing, information technology, education, and health sectors demonstrated resilience. Ding *et al.* (2022) analyze the effects of the public health crisis on the valuation of Chinese companies by investigating two key events during the pandemic. The first event they examine is the lockdown in Hubei province, and the second is

the decline in infection cases within China and its spread to other countries. They show that stock returns varied based on firms' network exposure to Hubei and foreign markets. They also find heterogeneity in the valuation effects of these exposures, which varied depending on a firm's ownership structure and industry.

Other papers have investigated the impact of COVID-19 on stock markets outside of China. Pandey and Kumari (2021) examine the responses of global stock markets, including both developed and emerging countries, to the initial outbreak of COVID-19 and early lockdown measures. Interestingly, they find that stock returns were significantly positive following early lockdowns and restrictions, suggesting that these measures boosted market confidence by curbing the spread of the virus. In contrast, Xie *et al.* (2022) reveal that the government announcements of lockdown generally had negative impacts on global stock markets using a sample of 44 countries. The response to the lifting announcement, however, was inconsistent. For example, during the event period (day 0–day 5), the cumulative abnormal returns of stock markets in eight countries were statistically significant and positive, while those in another eight countries were statistically significant and negative. This inconsistency makes it difficult to draw a clear conclusion about the effect of reopening on global stock reactions. Alam *et al.* (2021) focus on the Australian stock market's response to the official announcement of COVID-19 cases in Australia. They find that stock return reactions varied across industry sectors. Srinivasan *et al.* (2023) focus on the effect of social media communication, specifically Twitter, in preserving firms' stock returns during the first COVID-19 lockdown in the United States. Based on signaling theory, they examine how communication strategies such as message frequency and relevance of communication affect abnormal returns. They find that frequent and relevant positive messages helped sustain firms' stock prices during the pandemic.

While several studies have examined the impact of the COVID-19 pandemic on firms' financial performance, with many focusing on the Wuhan lockdown, one of the earliest measures implemented to contain the virus (He *et al.* 2020; Choudhury *et al.* 2022; Ding *et al.* 2022; Wang *et al.* 2022; Xia *et al.* 2023), the present study focuses on the Shanghai lockdown, which was announced on April 4, 2022, during a later phase of the pandemic. This period differed from the initial outbreak in several ways. According to WHO, by December 2020, the first COVID-19 vaccines had been administered, and over time, vaccine distribution expanded globally (World Health Organization NA). Also, the initial wave of infections subsided, and global economies began to rebound from the severe shocks of the early pandemic, entering a recovery phase (Park *et al.* 2024). This resulted in altering the public's perception of the pandemic, leading to optimism. However, the emergence of the highly infectious Omicron variant which occurred in late 2021 introduced new threats and challenges, leading to a resurgence of infection cases and requiring additional strict measures such as the Shanghai lockdown (Bradsher 2022). As our study examines the Shanghai lockdown during this later stage of the pandemic, we offer valuable insights into how these subsequent disruptions affected firms and their suppliers after they had already experienced earlier phases of the pandemic.

### 2.3. The Propagation Effect of COVID-19 on the Supply Chain

Several studies have examined the propagation effect of COVID-19 on supply chains by investigating how disruptions spread to a firm's customers and suppliers. Xia *et al.* (2023) examine the impact of supply chain disruptions caused by the Wuhan lockdown on the market values of Chinese firms. Their sample includes firms that are publicly traded on the Shanghai and Shenzhen exchanges and voluntarily disclose their top five suppliers in annual reports. They show that the sample firms experienced negative stock price changes from the Wuhan lockdown, and firms that have major suppliers in Wuhan experienced a worse market value loss compared to those without major suppliers in Wuhan. Wang *et al.* (2022) analyze the financial performance of 505 Chinese firms affected by COVID-19-related supply chain disruptions. They report immediate negative reactions in the Chinese stock market following the outbreak in China, with shareholder value declines for firms directly and indirectly impacted by COVID-19. Choudhury *et al.* (2022) examine 206 supply chain disruption events due to the pandemic between 2020 and 2021, estimating abnormal returns for 145 firms across 21 countries. They classify firms into supply-side or demand-side groups based on the type of disruption—for instance, shutdowns are supply-side disruptions, and price hikes are demand-side disruptions. They find that supply-side disruptions led to greater declines in shareholder value than demand-side disruptions. They investigate propagation effects based on firms' roles in supply chain networks, and they also show that firms with more complex supply chains were less affected by these disruptions.

Our study seeks to understand the impact of the Shanghai lockdown from a supply-side perspective by focusing on the international tier-1 suppliers of global automotive firms. We find that global automotive manufacturers were frequently mentioned in news reports as being particularly affected by supply chain disruptions caused by the Shanghai lockdown. Therefore, our chosen focal firms are international automakers, and we examine their international suppliers to assess how the Shanghai lockdown's impact propagated beyond China. Our study contributes to the literature by examining the cross-border propagation effects of supply chain disruptions during a major lockdown event. This enables us to offer a broader, global perspective.

## 3. Methodology

We employ an event study methodology to assess the impact of the Shanghai lockdown on stock market reactions. This approach has been widely used in previous literature to estimate the effect of specific events on shareholder wealth (Mackinlay 1997). The method is grounded in the efficient market hypothesis, which assumes market rationality and posits that the impact of an event is rapidly reflected in shareholder value (Mackinlay 1997). By calculating abnormal stock returns, the method captures deviations in stock prices from their expected trajectory in the absence of the event, using the stock price movements observed during



the pre-event estimation period. This approach allows us to isolate the causal relationship between the event and firms' financial performance while controlling for other factors that may influence stock prices (Brown and Warner 1985).

We calculate abnormal stock returns using an event study methodology to measure stock market reactions to both the Shanghai lockdown and its subsequent lifting. This section provides a rationale for selecting the event period and outlines the steps involved in calculating abnormal stock returns.

### 3.1. Event Period Selection

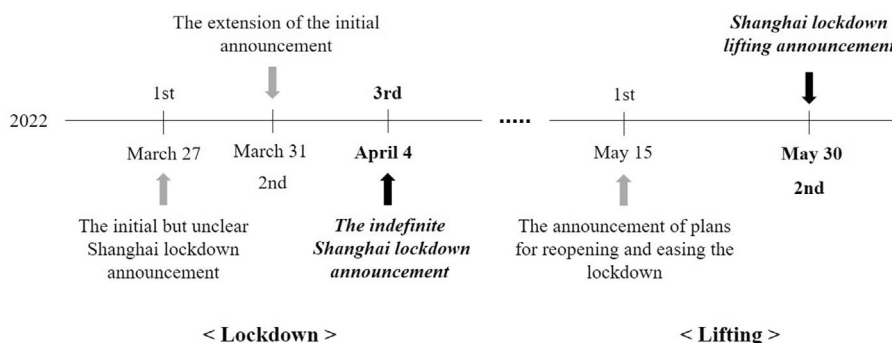
Following prior studies (He *et al.* 2020; Alam *et al.* 2021; Xia *et al.* 2023), we use the date of the government's official announcement of the lockdown as the event day. In the case of the Shanghai lockdown, the government made several announcements that reversed previous decisions before making the final announcement of an indefinite lockdown on April 4, 2022 (see Figure 1).

The first announcement regarding the Shanghai lockdown was made late in the evening on Sunday, March 27, 2022 (Jie *et al.* 2022; Kubota 2022; Mitchell *et al.* 2022). Authorities stated that a two-stage lockdown would be implemented starting the following day. The first stage was about imposing restrictions on the eastern region of the city from March 28 to April 1 (Jie *et al.* 2022; Kubota 2022; Mitchell *et al.* 2022). The second stage was planned to extend the lockdown to the western region for another 4 days starting after the eastern region's lockdown was lifted on April 1 (Jie *et al.* 2022; Kubota 2022; Mitchell *et al.* 2022). However, on the evening of March 31, 2022, authorities announced an extension of the lockdown in the eastern region while imposing the lockdown for the western region as planned (Hale *et al.* 2022a).

A few days later, authorities issued another announcement stating that the Shanghai lockdown would be extended with stricter restrictions, but this time without specifying when the lockdown would be lifted (Fan 2022). This announcement was made in the late evening of April 4, 2022, in China. As this marked a shift from a temporary measure to an indefinite one, it received much greater public attention than the earlier announcements. According to Google Trends and Baidu Search Index, the number of searches for the keyword "Shanghai lockdown" surged sharply after the last announcement (see Figures A.1 and A.3 in Appendix A). We therefore select the date of the last announcement, April 4, 2022, as our event day. We also conducted analyses using the earlier announcements on March 27, 2022, and March 31, 2022. However, these announcements did not significantly affect the stock prices of our sample firms as their abnormal stock returns were insignificant (see Tables B.1–B.2 in Appendix B). This implies that the initial and second lockdown announcements did not capture enough attention in the stock markets, thereby justifying the selection of the April 4 announcement date for our event study.

We then need to set day 0, the first trading day on which a respective stock market can react to the indefinite Shanghai lockdown announcement. Note that we consider tier-1 suppliers of four focal firms traded in different stock markets. Since



**Figure 1** Timeline of the Shanghai lockdown announcements and the lifting announcements

most North and South American and European stock markets were open when the indefinite lockdown announcement was released, the announcement date, April 4, becomes day 0. However, Asia-Pacific stock markets were closed at the time of the announcement, therefore April 5 became day 0 except for Taiwanese, Chinese, and Hong Kong stock markets, which were closed on April 5 due to national holidays. In this case, we set April 6 as day 0. Then, day 1 is the trading day after day 0, and day  $-1$  is the trading day before day 0 in respective countries.

Our study also examines the impact of the lockdown's lifting. After more than 1 month since the lockdown announcement, Shanghai officials revealed plans for reopening and easing the restrictions on the evening of May 15, 2022 (Kahn 2022). The following day, authorities stated that they would begin a steady reopening from June 1, 2022, if progress in controlling the pandemic was sustained (Hua 2022). The initial announcement suggested the possibility of easing restrictions, but it did not guarantee a definitive lifting of the lockdown. According to *Reuters*, however, on May 30, 2022, Shanghai authorities officially announced that the lockdown would be lifted as of June 1, 2022 (Goh and Woo 2022). Similar to the lockdown announcement, the last official lifting announcement gained the general public's attention more than earlier announcements (see Figures A.2 and A.4 in Appendix A), and the impact of the second lifting announcement was significantly greater than the first announcement (see Tables B.1–B.2 in Appendix B). Therefore, we select May 30, 2022, as the event day to study the effects of the lockdown lift (see Figure 1).

Since most North and South American and European stock markets were open on the lifting announcement date, May 30 is set as day 0 for the second event study, with one exception. US stock markets were closed on May 30 due to a national holiday, so May 31 is set as day 0 for the US stock market. Also, Asia-Pacific stock markets were possibly closed on May 30 when this news was released, meaning that May 31 is set as day 0 for these markets as well.

In event studies, it is common to use a one- or two-day event period to calculate abnormal stock returns, as all available information is typically quickly reflected in stock prices based on the efficient market hypothesis (Hendricks and Singhal 2003; Thirumalai and Sinha 2011; Lo *et al.* 2018). However, when an event has a prolonged effect on the market, a longer event period, such as 6 or 7 days, can be considered (Jacobs and Singhal 2020; Jacobs *et al.* 2022). Since we observed that public attention remained focused for several days following the indefinite lockdown announcement, we selected a 3-day event period (day 0 to day 2) to investigate stock market reactions to the lockdown.

Additionally, on April 11, 2022, Shanghai authorities announced plans to partially ease the strict lockdown restrictions due to increasing pressure from financial and social factors (Hale *et al.* 2022b). For firms listed in the Taiwanese, Chinese, and Hong Kong markets, April 11 is day 3. Including day 3 in the analysis by extending the event period to four or more days could introduce a confounding effect from this easing announcement, affecting the abnormal returns attributed to the lockdown announcement on April 4, 2022. This limits us to using the 3-day event period. Nevertheless, we provide the results based on longer event periods.

### 3.2. Estimation of Abnormal Stock Market Returns

To estimate abnormal returns, we employ the market model, which has been widely used in event studies (Chae *et al.* 2014; Han and Kwon 2015; Kodiyatt *et al.* 2024). The market model assumes that the stock return of a firm and the return of the market have a relationship over time:

$$R_{it} = \alpha_i + \beta_i R_{mt} + \varepsilon_{it} \quad (1)$$

where  $R_{it}$  is the stock return of firm  $i$  on day  $t$ ;  $R_{mt}$  is the market return of market  $m$  on day  $t$ ;  $\alpha_i$  is the intercept;  $\beta_i$  is the systemic risk that firm  $i$ 's stock has. In this model,  $\beta_i R_{mt}$  indicates the portion of firm  $i$ 's stock return by market movements;  $\varepsilon_{it}$  indicates the portion unexplained by market movements. Following previous event studies (e.g., Hendricks *et al.* 2020; Jacobs and Singhal 2020; Jacobs *et al.* 2022), we generated  $R_{mt}$  using a dominant market index of firm  $i$ 's country. For example, we employ the CRSP Value Weighted Index, Nikkei 225 Index, and CAC 40 Index for the US, Japanese, and French companies, respectively.

We utilize an ordinary least squares regression to estimate the regression coefficients  $\hat{\alpha}_i$  and  $\hat{\beta}_i$  over an estimation period before the event date. In previous literature, the estimation period is set as 200 trading days from day  $-210$  to day  $-11$  (Arora *et al.* 2020; Jacobs and Singhal 2020; Jacobs *et al.* 2022). However, in our case, before the last indefinite lockdown announcement, there had already been two announcements that may have influenced stock markets (see Figure 1). The first announcement (March 27, 2022) was made 8 days before our event day, that is, April 4, 2022. To eliminate the influence of those earlier announcements in calculating regression coefficients, we shift the estimation period back by 8 days. We also include only those firms

that have at least 40 stock return observations during the estimation period (Xia et al. 2016; Arora et al. 2020; Hendricks et al. 2020). The abnormal return for firm  $i$  on day  $t$ , denoted as  $A_{it}$ , is calculated by subtracting the expected return from the actual return where the expected return for firm  $i$  on day  $t$  is estimated as  $(\hat{\alpha}_i + \hat{\beta}_i R_{mt})$ :

$$A_{it} = R_{it} - (\hat{\alpha}_i + \hat{\beta}_i R_{mt}) \quad (2)$$

The mean abnormal return on day  $t$  is then calculated as;

$$\bar{A}_t = \sum_{i=1}^N \frac{A_{it}}{N} \quad (3)$$

where  $N$  is the number of sample supplier firms on day  $t$ .

We start calculating the abnormal return for sample firms over the same calendar day, and we compute the mean abnormal return  $\bar{A}$  over the 200-day estimation period as:

$$\bar{A} = \frac{\sum \bar{A}_t}{200} \quad (4)$$

Next, the standard deviation for daily mean abnormal returns over the 200-day estimation period is calculated as:

$$\hat{S}(\bar{A}_t) = \sqrt{\sum (\bar{A}_t - \bar{A})^2 / 199} \quad (5)$$

For day  $t$ , we generate the test statistic  $TS_t$  as:

$$TS_t = \bar{A}_t / \hat{S}(\bar{A}_t) \quad (6)$$

We use a  $t$ -test to statistically test the significance of the mean abnormal return on day  $t$ . The cumulative abnormal return (CAR) is computed for a given period  $(t_1, t_2)$  as:

$$CAR(t_1, t_2) = \sum_{t=t_1}^{t_2} \bar{A}_t \quad (7)$$

For a  $j$ -day event period, we generate the test statistic  $TS_j$  similar to a single day as:

$$TS_j = \frac{\sum_{t=t_1}^{t_2} \bar{A}_t}{\hat{S}(\bar{A}_t) \sqrt{j}} \quad (8)$$

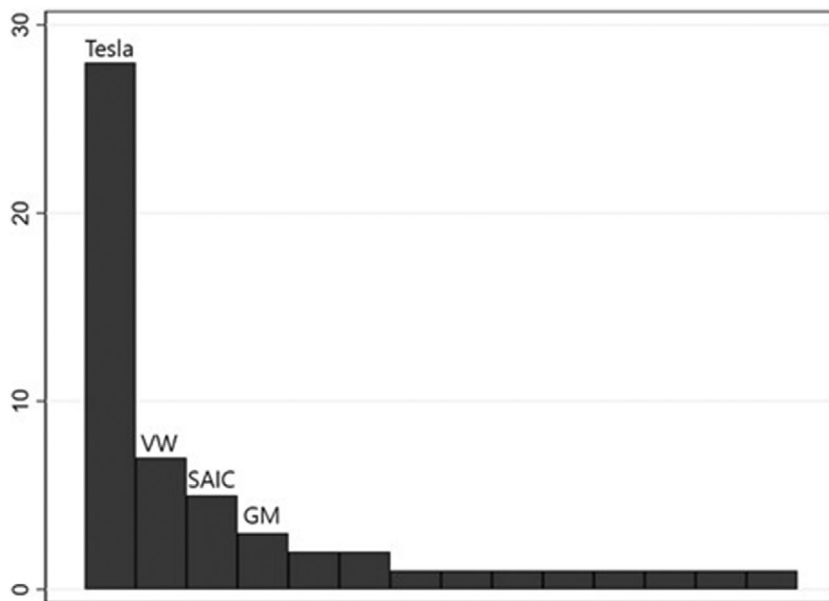
Since outliers can distort the mean, we also report the median and the percent of negative abnormal returns. We use the Wilcoxon signed-rank test to test the significance of the median abnormal return and the binomial sign test to test whether the percent of negative abnormal returns is significantly different from 50%.

#### 4. Sample Selection and Description

To select the focal firms for the event study, we followed the approach of Hendricks *et al.* (2020), using newspaper sources to identify the firms most frequently reported as being affected by the event. We searched *The Wall Street Journal*, *Financial Times*, and *Reuters* using the keyword “Shanghai lockdown” from March 27, 2022, to April 8, 2022. We identified the firms most frequently mentioned as being disrupted by the lockdown. This period covers from when the initial temporary lockdown announcement was made and 2 days<sup>1</sup> after the date when an indefinite lockdown announcement was made. We found 34 articles that directly reported supply chain disruptions (e.g., problems in operations or production halts) regarding the Shanghai lockdown. Figure 2 summarizes the frequency with which each firm appears in the news articles associated with supply chain disruptions from the Shanghai lockdown. Notably, the top four firms mentioned in the articles were all international automotive makers. Tesla was most frequently mentioned, more than six times compared to other firms, followed by Volkswagen, SAIC Motors, and General Motors (hereafter VW, SAIC, and GM). The news articles reported that the four firms’ factories in Shanghai were operating under a closed-loop system or halted production due to the lockdown measure. For example, Tesla shut down its plant in Shanghai after the initial lockdown announcement (Kubota 2022). VW’s joint ventures with SAIC initially ran their facilities in Shanghai under a closed-loop system but eventually halted production (Kubota and Wei 2022). As a firm’s negative media coverage is known to be associated with its stock market value (Xu *et al.* 2016; Jacobs and Singhal 2020; Jacobs *et al.* 2022), we select the top four mentioned firms—that is, Tesla, VW, SAIC, GM—as focal firms and examine the stock market reactions of these firms as well as their supply chain partners to the Shanghai lockdown and its lifting. Since the select firms are all automotive makers, their customers are limited to a few automobile dealerships. Thus, we focus

<sup>1</sup>As explained in Section 3, we set the date with an indefinite lockdown announcement as day 0 and consider a 3-day event period. We therefore searched the news articles until 2 days after the event day, day 0.

**Figure 2** The number of frequencies that each firm appears in news articles associated with supply chain disruptions from the Shanghai lockdown



on their tier-1 suppliers and investigate whether they experienced abnormal stock market reactions during the lockdown and its lifting periods.

We identify the tier-1 suppliers of the four focal firms using the Bloomberg Supply Chain (SPLC) database. This database provides information on the nature and scale of business relationships between focal firms and their suppliers and customers (Jacobs and Singhal 2020; Jacobs *et al.* 2022). We compiled the data from Bloomberg SPLC in June 2023, selecting tier-1 suppliers that maintained their relationships with the focal firms at the time of the lockdown announcement. In this database, suppliers are classified into four types: COGS, SG&A, CAPEX, and R&D. COGS indicates the Cost of Goods, which means the cost the buyer spends on materials and labor directly used in producing goods. SG&A indicates the cost a buyer pays for selling and delivering products and providing services directly and indirectly. CAPEX refers to capital expenditure, which indicates the expenditure the company pays for acquiring and maintaining physical assets. R&D implies that a supplier provides research and development resources to a buyer for innovation and improvements of the buyer's products. For this study, we restrict our sample to COGS suppliers, which provide direct materials to the focal firms (Jacobs *et al.* 2022). Also, Jacobs and Singhal (2020) report that only COGS suppliers were

**Table 1** The number of suppliers for the four focal firms

The total number of sample suppliers of the four focal firms is 252. The number of suppliers contracting more than one focal buyer firm is 146.

|                    | Tesla | Volkswagen | SAIC Motors | General Motors |
|--------------------|-------|------------|-------------|----------------|
| Shared suppliers   | 70    | 133        | 45          | 140            |
| Distinct suppliers | 15    | 32         | 7           | 52             |
| <i>N</i>           | 85    | 165        | 52          | 192            |

significantly affected by the Volkswagen scandal compared to SG&A and CAPEX suppliers. We exclude firms in Saudi Arabia due to differing trading dates from other stock markets and remove those whose GVKEY, a proprietary identifier in Compustat database, is not observed in Compustat Global and North America data from 2020 to 2022 and those whose main financial values are missing. Figure A5 in Appendix A provides the process of constructing the final sample.

The aggregated number of COGS tier-1 suppliers of the four focal firms is 252. Table 1 summarizes the number of suppliers for the four focal firms. Panel A and Panel B of Table 2 present the country distribution and sample statistics of these suppliers. We also find that most suppliers were in the four industry sectors—motor vehicle parts, tires, electronic components, and semiconductors and related devices (see Panel C of Table 2).

To set a benchmark against the focal firms, we select a set of comparable automobile companies that do not have manufacturing plants in Shanghai. We anticipate that these benchmark firms and their suppliers will be less affected by the Shanghai lockdown compared to the focal firms and their suppliers. We first check the ranking of international automakers by revenue for the year preceding the lockdown (i.e., 2021) using GlobalData and Bloomberg, focusing on those ranked in the top 10. After excluding the focal firms within this group, we verify whether each remaining firm operates car manufacturing plants in Shanghai. While Mercedes-Benz Group and Honda Motor Company had plants in Shanghai at the time of the lockdown, we identify five companies—Toyota, Stellantis, Ford Motor Company, Bayerische Motoren Werke Group AG, and Hyundai Motor Company—that did not operate plants in Shanghai. We select these five companies as our benchmark.

To ensure a reliable comparison between the focal and benchmark firms, we compare their sales, total assets, and market value, confirming that there are no statistically significant differences (see Table 3). The suppliers of the benchmark firms are selected using the same procedure of selecting the suppliers of the focal firms. The total number of suppliers for the benchmark firms is 354, of which 196 suppliers are common to both the focal and benchmark firms (see Figure 3). Therefore,

**Table 2** Sample statistics of the four focal firms' suppliers

Panel B shows the sample statistics of suppliers. Total assets and sales are values at the end of the fiscal year before March 2022, and market value is the value 8 days before the initial lockdown announcement on March 27, 2022.

| Panel A: Countries of suppliers |                             |           |                             |
|---------------------------------|-----------------------------|-----------|-----------------------------|
| Countries                       | The number of suppliers (%) | Countries | The number of suppliers (%) |
| Japan                           | 71 (28.17%)                 | China     | 12 (4.76%)                  |
| USA                             | 71 (28.17%)                 | Canada    | 6 (2.38%)                   |
| Korea                           | 18 (7.14%)                  | Hongkong  | 6 (2.38%)                   |
| Germany                         | 17 (6.75%)                  | India     | 6 (2.38%)                   |
| France                          | 12 (4.76%)                  | Others    | 33 (13.11%)                 |

| Panel B: Sample statistics of suppliers |           |          |           |
|-----------------------------------------|-----------|----------|-----------|
|                                         | Mean      | Median   | Std. Dev  |
| Total assets (\$m)                      | 14 453.72 | 4778.313 | 32 764.48 |
| Sales (\$m)                             | 11 512.4  | 3460.514 | 24 687.05 |
| Market value (\$m)                      | 11 396.63 | 2526.927 | 29 966.39 |

| Panel C: Industry of suppliers               |               |
|----------------------------------------------|---------------|
| SIC code                                     | Frequency (%) |
| 3714 – Motor vehicle parts and accessories   | 71 (28.17%)   |
| 3011 – Tires and inner tubes                 | 11 (4.37%)    |
| 3670 – Electronic components and accessories | 11 (4.37%)    |
| 3674 – Semiconductors and related devices    | 10 (3.97%)    |
| Other industries                             | 149 (59.12%)  |

in analyses comparing the suppliers of focal and benchmark firms, we exclude these common suppliers and focus on those unique to each set.

## 5. Results

### 5.1. Stock Market Reaction for Focal Firms

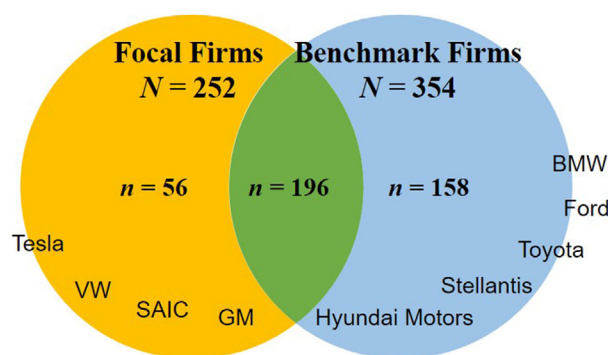
We begin by analyzing the abnormal returns for the four focal firms. Table 4 shows the aggregated abnormal stock returns for all focal firms in response to the lockdown and lifting announcements. Figure 4 provides line plots of the CARs from day −5 to day 6 (see Figure A6 in Appendix A for all announcements). The aggregated abnormal returns are calculated over a 3-day event period (i.e., day 0–day 2) and represent our main findings, as explained before. Notably, the results indicate that the abnormal returns for all focal firms were insignificant for both the lockdown and the lifting events.



**Table 3** Sample statistics of the four focal firms and the benchmark firms

Total assets and sales are values at the end of the fiscal year before March 2022, and market value is the value 8 days before the initial lockdown announcement on March 27, 2022.

|                           | Sales (\$m) | Total assets (\$m) | Market value (\$m) |
|---------------------------|-------------|--------------------|--------------------|
| Panel A: Four focal firms |             |                    |                    |
| Tesla                     | 53 823      | 62 131             | 822 000            |
| VW                        | 295 928     | 601 874            | 97 712             |
| SAIC                      | 120 918     | 144 259            | 33 082             |
| GM                        | 127 004     | 244 718            | 60 315             |
| Panel B: Benchmark firms  |             |                    |                    |
| BMW                       | 131 570     | 261 339            | 47 389             |
| Stellantis                | 176 728     | 195 573            | 45 705             |
| Toyota                    | 256 718     | 562 943            | 258 993            |
| Ford                      | 136 341     | 257 035            | 64 228             |
| Hyundai Motors            | 102 765     | 196 595            | 29 207             |

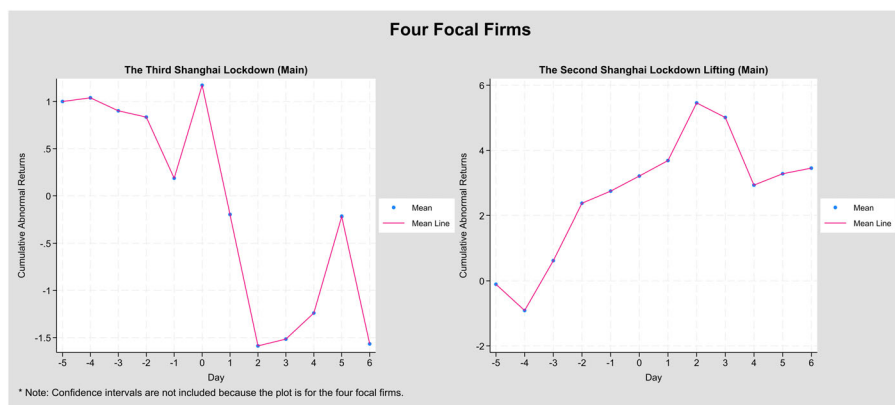
**Figure 3** Diagram of suppliers of the focal firms and the benchmark firms

We propose two possible explanations for this finding. Previous research by Hendricks and Singhal (2003) demonstrates that larger firms are better positioned to mitigate the negative impact of disruptive events. Given that the focal firms are larger than their suppliers, it is possible that they were better equipped to manage the supply chain disruptions caused by the lockdown. Furthermore, the Shanghai lockdown was not an event specifically targeting a single company, unlike the Volkswagen emissions scandal (Jacobs and Singhal 2020) and the 2018 US government ban on ZTE (Jacobs *et al.* 2022), which triggered significant stock market

**Table 4** Aggregated abnormal returns of the four focal firms for the main announcements ( $N = 4$ )

<sup>a</sup>Obtained from Wilcoxon signed-rank tests. <sup>b</sup>Obtained from binomial sign tests. \* $p < 0.1$ , \*\* $p < 0.05$ , \*\*\* $p < 0.01$ .

| Event day(s)                                           | Mean   | $t$   | Median | $Z^a$  | % Negative | $Z^b$  |
|--------------------------------------------------------|--------|-------|--------|--------|------------|--------|
| Panel A: Lockdown announcement on April 4, 2022 (Main) |        |       |        |        |            |        |
| 0                                                      | 0.99%  | 0.86  | 0.34%  | 0.73   | 50.00%     | 0.00   |
| 1                                                      | -1.37% | -1.19 | -1.81% | -1.46  | 75.00%     | 1.00   |
| 2                                                      | -1.39% | -1.21 | -1.59% | -1.83* | 100.00%    | 2.00** |
| (-5, 0)                                                | 1.17%  | 0.42  | 2.06%  | 0.73   | 25.00%     | -1.00  |
| (0, 1)                                                 | -0.38% | -0.24 | -0.29% | -0.73  | 50.00%     | 0.00   |
| (0, 2)                                                 | -1.78% | -0.89 | -1.71% | -0.73  | 50.00%     | 0.00   |
| (0, 6)                                                 | -1.75% | -0.58 | -0.29% | -0.37  | 50.00%     | 0.00   |
| Panel B: Lifting announcement on May 30, 2022 (Main)   |        |       |        |        |            |        |
| 0                                                      | 0.46%  | 0.40  | -0.02% | 0.00   | 50.00%     | 0.00   |
| 1                                                      | 0.48%  | 0.42  | 0.01%  | 0.37   | 50.00%     | 0.00   |
| 2                                                      | 1.77%  | 1.54  | 1.73%  | 1.46   | 25.00%     | -1.00  |
| (-5, 0)                                                | 3.21%  | 1.14  | 5.23%  | 1.10   | 25.00%     | -1.00  |
| (0, 1)                                                 | 0.94%  | 0.58  | -0.31% | -0.37  | 75.00%     | 1.00   |
| (0, 2)                                                 | 2.71%  | 1.37  | 0.43%  | 0.73   | 50.00%     | 0.00   |
| (0, 6)                                                 | 0.71%  | 0.23  | -0.23% | 0.00   | 50.00%     | 0.00   |

**Figure 4** Line plots of CARs of the focal firms for the main announcements

reactions for specific firms. As a result, the non-significant stock market reaction for the focal firms can be attributed to the nature of the event that is not specific to a single firm and the limited sample size of focal firms (i.e.,  $n = 4$ ).

**Table 5** Aggregated abnormal returns of the suppliers of the four focal firms for the main announcements ( $N = 252$ )

<sup>a</sup>Obtained from Wilcoxon signed-rank tests. <sup>b</sup>Obtained from binomial sign tests. \* $p < 0.1$ , \*\* $p < 0.05$ , \*\*\* $p < 0.01$ .

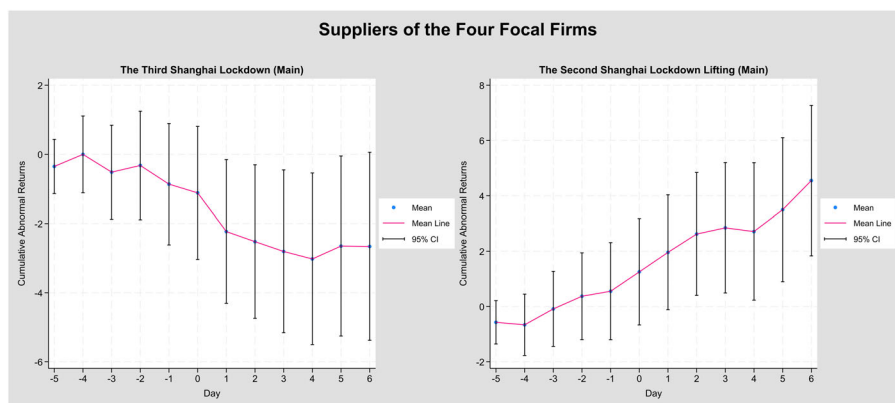
| Event day(s)                                           | Mean   | $t$      | Median | $Z^a$    | % Negative | $Z^b$    |
|--------------------------------------------------------|--------|----------|--------|----------|------------|----------|
| Panel A: Lockdown announcement on April 4, 2022 (Main) |        |          |        |          |            |          |
| 0                                                      | -0.25% | -0.62    | -0.22% | -3.39*** | 58.73%     | 2.77***  |
| 1                                                      | -1.12% | -2.79*** | -0.92% | -9.02*** | 75.79%     | 8.19***  |
| 2                                                      | -0.29% | -0.72    | -0.14% | -2.48**  | 55.56%     | 1.76*    |
| (-5, 0)                                                | -1.11% | -1.13    | -0.72% | -3.69*** | 60.71%     | 3.40***  |
| (0, 1)                                                 | -1.37% | -2.42**  | -1.35% | -7.80*** | 74.60%     | 7.81***  |
| (0, 2)                                                 | -1.66% | -2.39**  | -1.52% | -7.32*** | 69.05%     | 6.05***  |
| (0, 6)                                                 | -1.83% | -1.73*   | -1.45% | -5.39*** | 66.27%     | 5.17***  |
| Panel B: Lifting announcement on May 30, 2022 (Main)   |        |          |        |          |            |          |
| 0                                                      | 0.71%  | 1.77*    | 0.57%  | 5.80***  | 34.13%     | -5.04*** |
| 1                                                      | 0.69%  | 1.72*    | 0.71%  | 5.15***  | 34.92%     | -4.79*** |
| 2                                                      | 0.67%  | 1.67*    | 0.45%  | 5.36***  | 35.32%     | -4.66*** |
| (-5, 0)                                                | 1.26%  | 1.28     | 0.76%  | 4.07***  | 39.68%     | -3.28*** |
| (0, 1)                                                 | 1.40%  | 2.47**   | 1.29%  | 7.77***  | 28.17%     | -6.93*** |
| (0, 2)                                                 | 2.05%  | 2.96***  | 1.94%  | 9.03***  | 23.81%     | -8.32*** |
| (0, 6)                                                 | 3.20%  | 3.02***  | 2.52%  | 8.72***  | 23.41%     | -8.44*** |

## 5.2. Stock Market Reaction for Focal Firms' Suppliers

Next, we show the aggregated results for the suppliers of the focal firms concerning the lockdown and lifting announcements. Table 5 reports the daily CARs for all suppliers of the focal firms, and Figure 5 shows the line plots of the CARs with their 95% confidence intervals (see Figure A7 in Appendix A for all announcements). We find significant stock market reactions to both events when considering the aggregated sample of suppliers. For the lockdown announcement, the mean abnormal return on day 1 is significant at the 1% level, and mean CAR (0, 1) and CAR (0, 2) are also significant at the 5% level. The mean abnormal returns and CARs associated with the lifting announcement consistently show significantly positive values.

We compare the magnitude of the impact of the lockdown with that of the lifting. During the 3 days following the lockdown announcement, the focal firms' suppliers experienced an average decline in shareholder value of 1.66% ( $p = 0.017$ ). Following the lifting of the lockdown announcement, the same sample experienced an average increase of 2.05% ( $p = 0.003$ ) in their stock prices over the same period. We compare the size between the negative CAR (0, 2) of the lockdown and the positive CAR (0, 2) of the lifting, and there is no significant difference. These results suggest that the stock market value of suppliers varied to a similar degree as the situation of their focal firm shifted from disruption to subsequent recovery.

**Figure 5** Line plots of CARs of the suppliers of the four focal firms for the main announcements



### 5.3. The Effect of Suppliers' Characteristics on Stock Market Reaction

In this section, we use a regression model to examine whether the stock market reactions vary based on suppliers' characteristics. The dependent variable is the CAR for a 3-day event window, CAR (0, 2). We perform separate analyses for two distinct events: the lockdown announcement on April 4, 2022, and its subsequent lifting announcement on May 30, 2022.

The regression model incorporates supplier characteristics that reflect their capital and operating performance using two variables, *Capital Expenditures-to-Lagged Assets* and *Inventory Turnover Ratio*, and other characteristics—*R&D Intensity* and *Market Power*. *Capital Expenditures-to-Lagged Assets* measures the proportion of capital expenditures relative to assets from the previous year. It is calculated as the ratio of a firm's capital expenditure in 2021 to its total assets in 2020 (Aral et al. 2020). *Inventory Turnover Ratio* assesses operational performance efficiency by indicating how quickly inventory is converted to sales (Chen et al. 2023). *Inventory Turnover Ratio* is computed as the ratio of sales over the past 12 months to average inventory. *R&D Intensity* indicates R&D expenditures relative to sales for 2021 (Xia et al. 2016; Chang et al. 2022). Following Chang et al. (2022), we replaced all missing values of *R&D Intensity* with zero. *Market Power* reflects a supplier's market share, calculated as the ratio of the supplier's sales revenue to the total sales revenue of the industry to which a supplier belongs based on a two-digit SIC code (Chang et al. 2022).

We include several firm-level control factors that may influence the abnormal returns in the regression analysis—*Size*, *Leverage*, and *Market-to-Book*. *Size* indicates the firm size and is measured as the natural log of the firm's total assets in 2021 (Klößner et al. 2023). We expect larger firms will be better equipped to withstand

negative shocks compared to smaller firms, following Jacobs and Singhal (2020) and Jacobs *et al.* (2022). *Leverage* indicates financial leverage, which may affect shareholders' and bondholders' reactions differently. We computed *Leverage* as the ratio of total debt to total assets for 2021 (Jiang *et al.* 2023). *Market-to-Book* (*M\_B*) reflects the firm's growth potential and has been shown to influence stock market reactions (Jacobs and Singhal 2020; Jacobs *et al.* 2022). This ratio is operationalized as the market value of equity divided by the book value of equity for 2021.

Also, we included control variables to account for the effects of suppliers' focal buyer companies, industry sectors, and locations. Dummy variables *Tesla*, *VW*, *SAIC*, and *GM* show the supplier's focal buyer firms. *Industry\_3714*, *Industry\_3011*, *Industry\_3670*, and *Industry\_3674* are dummy variables indicating whether the supplier is included in the industry of motor vehicle parts and accessories, tires and inner tubes, electronic components and accessories, or semiconductors and related devices, respectively. The four industries are the most frequent industries to which suppliers belong (see Panel C of Table 2). Dummy variables *America* and *Asia-Pacific* indicate geographical areas where suppliers are located (Jacobs and Singhal 2020). We employ robust standard errors in our regression analysis to account for potential heteroskedasticity.

As shown in Table 6, our results indicate that during the lockdown event, a high *Inventory Turnover Ratio* led to more negative stock return reactions. In normal conditions, a high *Inventory Turnover Ratio* suggests that a firm's inventory is rapidly converted to sales or that suppliers maintain a lean system in managing their inventories, reflecting efficient inventory management. However, during the lockdown, this could be interpreted in two ways. First, suppliers with a high *Inventory Turnover Ratio* may have been more adversely affected because their inability to sell inventory during the lockdown led to greater losses. Second, firms employing a lean inventory system, which minimizes inventory levels, might lack sufficient materials and parts to sustain production during disruptions. This lean approach could cause production interruptions due to insufficient inventory redundancy, limiting flexibility in the face of supply chain disruptions (Choudhury *et al.* 2022).

*Capital Expenditure-to-Lagged Assets* also had a negative impact on stock returns during the lockdown. This implies that suppliers who had invested a significant amount of capital compared to their total assets prior to the lockdown faced more severe negative reactions from investors. We expect that these suppliers could not generate returns on their investments due to halted operations, leading to a more pronounced negative response to the lockdown announcement.

In the case of the lockdown lifting, we find that *Leverage* is statistically significant. *Leverage* shows a significantly positive effect on stock returns following the lifting of the lockdown. Higher *Leverage* indicates that a firm is operating with greater financial risk; thus, firms with higher *Leverage* might exhibit a more positive reaction to the lifting event, as reduced uncertainty can enhance their financial outlook. *R&D Intensity*, on the other hand, is statistically negative in the lifting event.

**Table 6** Regression results for the suppliers of the four focal firms; dependent variable is CAR (0, 2); lockdown announcement on April 4, 2022; lifting announcement on May 30, 2022Robust *t*-statistics in parentheses. The referent category is Europe. \* $p < 0.1$ , \*\* $p < 0.05$ , \*\*\* $p < 0.01$ .

| CAR (0, 2)                                   | (1)<br>Lockdown    | (2)<br>Lifting     |
|----------------------------------------------|--------------------|--------------------|
| <i>Size</i>                                  | 0.0489 (0.303)     | −0.0603 (−0.436)   |
| <i>Leverage</i>                              | −2.452 (−1.119)    | 2.240 (1.734)*     |
| <i>Market_to_Book</i>                        | 0.265 (1.650)      | 0.0703 (0.703)     |
| <i>R&amp;D Intensity</i>                     | −0.525 (−0.435)    | −2.420 (−2.361)**  |
| <i>Market Power</i>                          | 3.350 (1.091)      | 0.679 (0.378)      |
| <i>Capital Expenditures_to_Lagged Assets</i> | −14.15 (−2.220)**  | 3.169 (0.649)      |
| <i>Inventory_Turnover_Ratio</i>              | −0.0126 (−2.439)** | −0.000734 (−0.170) |
| <i>Tesla</i>                                 | 0.100 (0.200)      | 0.00872 (0.0197)   |
| <i>VW</i>                                    | 0.0169 (0.0393)    | 0.729 (1.738)*     |
| <i>SAIC</i>                                  | −0.802 (−1.419)    | 0.168 (0.327)      |
| <i>GM</i>                                    | 0.297 (0.528)      | 0.802 (1.747)*     |
| <i>Industry_3714</i>                         | −0.275 (−0.510)    | 2.464 (4.989)***   |
| <i>Industry_3011</i>                         | −0.314 (−0.412)    | 0.385 (0.373)      |
| <i>Industry_3670</i>                         | −0.405 (−0.554)    | 0.942 (1.074)      |
| <i>Industry_3674</i>                         | −1.403 (−1.671)*   | 0.710 (0.939)      |
| <i>America</i>                               | 0.204 (0.265)      | −0.631 (−0.980)    |
| <i>Asia-Pacific</i>                          | 1.603 (2.217)**    | −0.147 (−0.246)    |
| Constant                                     | −1.883 (−1.072)    | 0.139 (0.0930)     |
| Observations                                 | 241                | 241                |
| <i>R</i> -squared                            | 0.110              | 0.193              |
| <i>F</i> -value                              | 2.202***           | 2.893***           |

This finding suggests investors might expect that companies with high *R&D Intensity* could not fully utilize the opportunity to produce or sell their products after the lifting of the lockdown.

#### 5.4. Comparison Between the Suppliers of Focal and Benchmark Firms

In this section, we compare the CARs (0, 2) of the suppliers of the focal firms and benchmark firms in reaction to the lockdown and lifting announcements. This comparison tests whether investors for the supplier firms responded differently depending on whether their customer automotive firms operated car plants in Shanghai. As noted earlier, we exclude suppliers that appear common to both focal and benchmark groups, resulting in a final sample size of 56 suppliers for the focal group and 158 suppliers for the benchmark group.

For the lockdown event, the mean of CAR (0, 2) for the focal group was significantly negative at the 1% significance level (see Appendix B; Table B3), while the

mean of CAR (0, 2) of the benchmark group was also significantly negative, though only at the 10% significance level (see Suppliers of the benchmark firms' Panel A of Table 7). When comparing the CARs (0, 2) of the two groups, the focal group experienced significantly more negative cumulative stock return changes than the benchmark group ( $p = 0.014$ , one tail). This suggests that while both groups of suppliers received negative reactions from investors following the announcement of the indefinite lockdown in Shanghai, the magnitude of the negative impact was significantly greater for the focal group. This difference may be attributed to the fact that the focal firms operated plants in Shanghai, emphasizing the importance of the physical location of operations in determining the severity of supply chain disruptions (Choudhury *et al.* 2022).

In contrast, for the lifting announcement, there was no statistically significant difference in CARs between the two groups. This indicates that the positive market reaction to the lifting event impacted both the focal and benchmark groups to a similar degree, suggesting that investors viewed the lifting of restrictions as beneficial to the global automotive supply chain, regardless of the firms' specific operational locations.

## 6. Conclusion

An indefinite Shanghai lockdown was announced in the late evening of April 4, 2022, in response to a surge in coronavirus infections in the region. Given Shanghai's vital position in the global economy and supply chain networks, the stringent restriction imposed by the lockdown significantly affected both local and global businesses operating in Shanghai, as well as their supply chain partners. This study employs an event study methodology and investigates the stock market reactions to the Shanghai lockdown and its lifting. Specifically, we identify four focal firms—Tesla, Volkswagen, SAIC Motors, and General Motors—that were most frequently mentioned in the media as experiencing a supply chain disruption from the Shanghai lockdown. We also consider the tier-1 suppliers of these focal firms to understand whether the disruptive event also impacted their global suppliers.

We estimate stock market reactions for the focal firms and their suppliers by calculating abnormal returns. We find the abnormal returns were insignificant for the four focal firms we considered. The lockdown was not a firm-specific event and therefore the size of the impact was not significant at an individual firm level. However, we find significant stock market reactions aggregated over the sample of suppliers of the focal firms. When the indefinite lockdown announcement was made, the reaction was significantly negative, with an average decline of 1.66%. In contrast, when the lifting of the lockdown was announced, we found the opposite effect with a larger effect size. In this case, the same sample experienced an average increase of 2.05% in their stock prices. Overall, the findings indicate that the stock market value of suppliers fluctuates systematically as their focal firm moves from disruption to recovery.



**Table 7** Summary of aggregated abnormal returns of the benchmark firms and 158 suppliers only included in benchmark firms' suppliers for all announcements

<sup>a</sup>Obtained from Wilcoxon signed-rank tests. <sup>b</sup>Obtained from binomial sign tests. \* $p < 0.1$ , \*\* $p < 0.05$ , \*\*\* $p < 0.01$ .

| Benchmark firms ( $N = 5$ )                            |        |          |        |          |            |          |
|--------------------------------------------------------|--------|----------|--------|----------|------------|----------|
| Event day(s)                                           | Mean   | $t$      | Median | $Z^a$    | % Negative | $Z^b$    |
| Panel A: Lockdown announcement on April 4, 2022 (main) |        |          |        |          |            |          |
| 0                                                      | 0.31%  | 0.41     | 0.40%  | 1.21     | 40.00%     | -0.45    |
| 1                                                      | -0.96% | -1.26    | -1.30% | -0.67    | 60.00%     | 0.45     |
| 2                                                      | 0.17%  | 0.23     | 0.52%  | 0.41     | 40.00%     | -0.45    |
| (-5, 0)                                                | -1.71% | -0.92    | -2.73% | -1.21    | 60.00%     | 0.45     |
| (0, 1)                                                 | -0.65% | -0.61    | -0.80% | -0.67    | 60.00%     | 0.45     |
| (0, 2)                                                 | -0.48% | -0.36    | 0.01%  | 0.67     | 20.00%     | -1.34    |
| (0, 6)                                                 | 1.05%  | 0.52     | 1.05%  | 0.45     | 20.00%     | -1.34    |
| Panel B: Lifting announcement on May 30, 2022 (main)   |        |          |        |          |            |          |
| 0                                                      | 0.43%  | 0.57     | 0.49%  | 1.75*    | 20.00%     | -1.34    |
| 1                                                      | -0.44% | -0.58    | -0.99% | -0.41    | 60.00%     | 0.45     |
| 2                                                      | 0.93%  | 1.22     | 1.20%  | 1.48     | 20.00%     | -1.34    |
| (-5, 0)                                                | 5.08%  | 2.73***  | 6.41%  | 2.02**   | 00.00%     | -2.24**  |
| (0, 1)                                                 | -0.01% | -0.01    | -0.46% | -0.14    | 60.00%     | 0.45     |
| (0, 2)                                                 | 0.92%  | 0.70     | 0.50%  | 1.21     | 40.00%     | -0.45    |
| (0, 6)                                                 | -0.30% | -0.15    | -0.40% | -0.67    | 80.00%     | 1.34     |
| Suppliers of the benchmark firms ( $N = 158$ )         |        |          |        |          |            |          |
| Event day(s)                                           | Mean   | $t$      | Median | $Z^a$    | % Negative | $Z^b$    |
| Panel A: Lockdown announcement on April 4, 2022 (main) |        |          |        |          |            |          |
| 0                                                      | 0.10%  | 0.26     | 0.01%  | -0.09    | 48.10%     | -0.48    |
| 1                                                      | -0.92% | -2.44*** | -0.79% | -6.76*** | 72.78%     | 5.73***  |
| 2                                                      | -0.42% | -1.11    | -0.32% | -3.39*** | 61.39%     | 2.86***  |
| (-5, 0)                                                | 0.51%  | 0.41     | 0.03%  | 0.42     | 47.47%     | -0.64    |
| (0, 1)                                                 | -0.83% | -1.55    | -0.76% | -4.13*** | 62.66%     | 3.18***  |
| (0, 2)                                                 | -1.28% | -1.96*   | -1.00% | -5.27*** | 70.25%     | 5.09***  |
| (0, 6)                                                 | -1.13% | -1.13    | -0.71% | -2.63*** | 49.37%     | -0.16    |
| Panel B: Lifting announcement on May 30, 2022 (main)   |        |          |        |          |            |          |
| 0                                                      | -0.18% | -0.48    | -0.16% | -1.41    | 55.06%     | 1.27     |
| 1                                                      | 0.96%  | 2.55**   | 0.90%  | 4.82***  | 30.38%     | -4.93*** |
| 2                                                      | 0.54%  | 1.43     | 0.14%  | 2.81***  | 42.41%     | -1.90*   |
| (-5, 0)                                                | 1.21%  | 1.30     | 1.32%  | 3.19***  | 34.81%     | -3.82*** |
| (0, 1)                                                 | 0.74%  | 1.38     | 0.47%  | 2.85***  | 35.44%     | -3.66*** |
| (0, 2)                                                 | 1.29%  | 1.98**   | 0.90%  | 5.07***  | 33.54%     | -4.14*** |
| (0, 6)                                                 | -0.32% | -0.32    | 0.11%  | -0.30    | 45.57%     | -1.11    |

Our analysis reveals that the impact of the lockdown differed depending on supplier characteristics, such as capital investment and operational performance. We found that suppliers who had higher capital expenditures in 2021 relative to their 2020 total assets were more significantly impacted by the supply chain disruptions than those with lower expenditures. Suppliers that rapidly converted inventory to sales before the lockdown experienced more negative stock returns than those that did not. Moreover, a supplier's financial profile and innovation effort played an important role in determining the impact of the lifting announcement. Investors reacted more positively to the lifting event for companies that operated with higher risk, while those with high R&D spending over their sales were negatively affected by the event.

We also compare the influence of the geographical location of a firm's manufacturing plant on stock market reactions. By comparing the cumulative abnormal stock returns of the 56 suppliers of the focal firms with those of 158 suppliers from the benchmark group, we find that the suppliers of the focal firms were more negatively impacted by the lockdown announcement than those in the benchmark group. However, for the lifting announcement, the difference between the two groups was not statistically significant. This indicates that while the physical location of operations plays a critical role in the severity of supply chain disruptions, it is less relevant to the recovery from such disruptions.

Although we did not directly compare our sample with that of Wang *et al.* (2022) to analyze the effect of the pandemic on supply chain disruptions over time, we find that mean and median CAR (0, 2) for the Shanghai lockdown (−1.66% and −1.52, respectively) are lower in magnitude than those reported by Wang *et al.* (2022) for COVID-19 in 2020 (−2.67% and −4.17%, respectively) for Chinese firms. This suggests that the Shanghai lockdown in 2022 may have had a weaker impact than the initial disruption in 2020. One possible explanation is that stock market reactions were more pronounced in the early stages of the pandemic, when the impact was highly uncertain, but became less severe over time, even in response to a full-scale city lockdown.

In this paper, we focus on the supply chains of the focal firms identified in news articles as being most heavily impacted by the Shanghai lockdown. Since these focal firms were all automakers with few automobile dealerships as customers, examining the downstream propagation of supply chain disruptions was not feasible in our study. However, prior research has shown the existence of downstream propagation effects from supply chain disruptions (Hendricks *et al.* 2020; Jacobs and Singh 2020; Jacobs *et al.* 2022; Wang *et al.* 2022). Therefore, future research could explore stock market reactions for customers of other focal firms during the pandemic, providing valuable insights into the broader impact of pandemic-induced disruptions and recovery.

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Appendix A

Figure A1 The number of searches related to the Shanghai lockdown using Google Trends

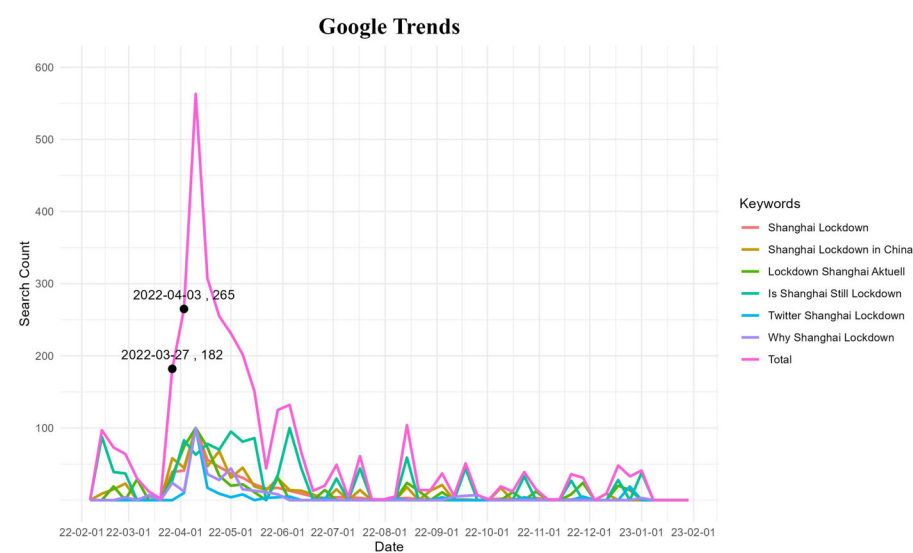
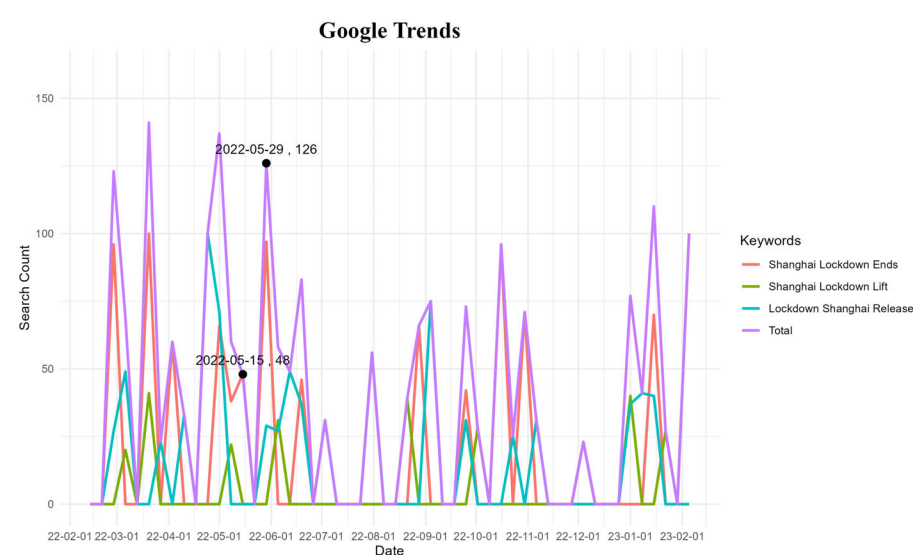
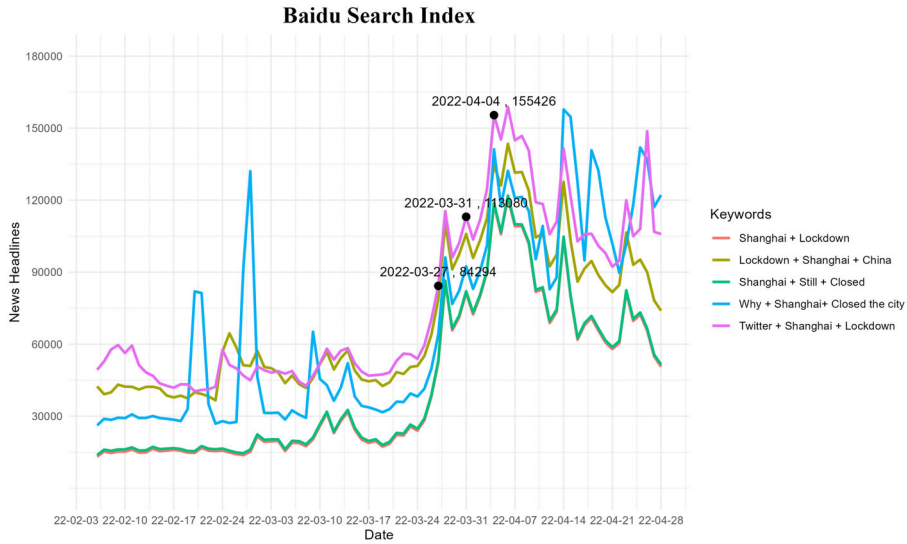


Figure A2 The number of searches related to the Shanghai lockdown lifting using Google Trends



**Figure A3** The number of searches related to the Shanghai lockdown using Baidu Search Index



**Figure A4** The number of searches related to the Shanghai lockdown lifting using Baidu Search Index

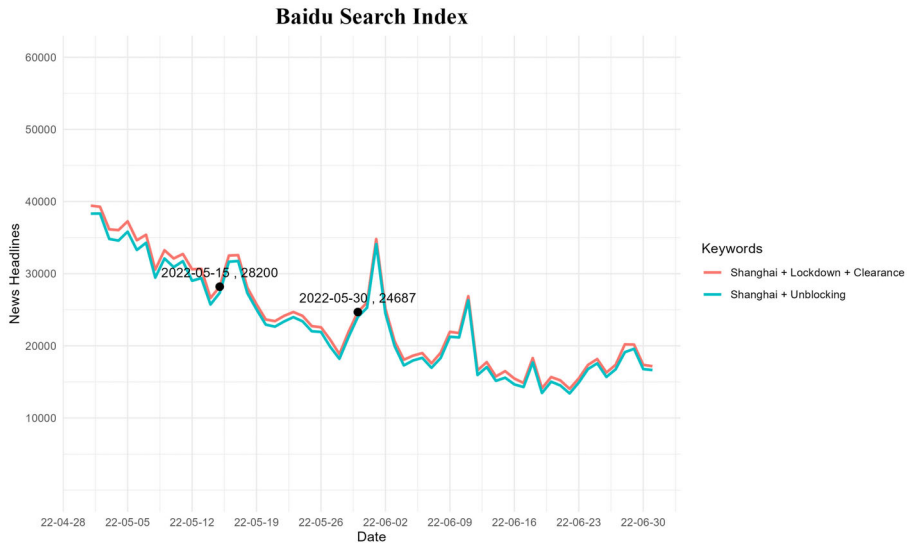




Figure A5 The process of constructing final sample suppliers of the four focal firms

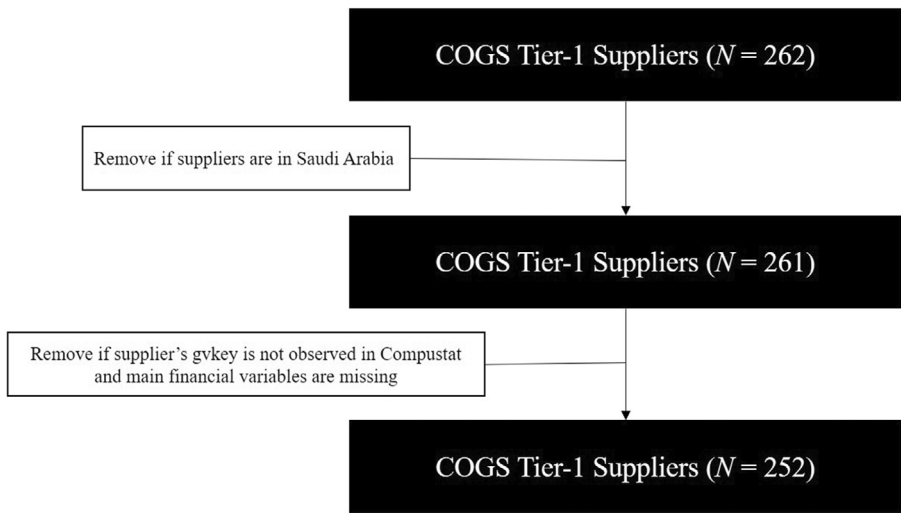


Figure A6 Line plots of CARs of the four focal firms for all announcements

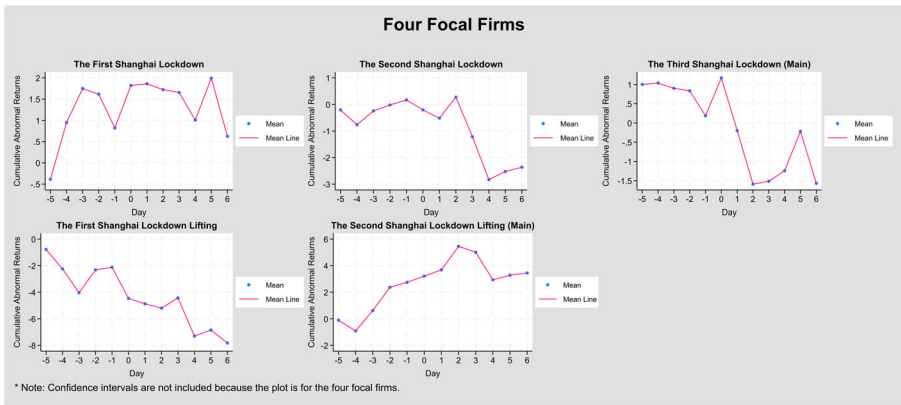


Figure A7 Line plots of CARs of the suppliers of the four focal firms for all announcements



## Appendix B

**Table B1** Aggregated abnormal returns for the four focal firms for all announcements ( $N = 4$ )

This table shows all the event study results for the five announcements. Panel A indicates the result of the initial but unclear Shanghai lockdown announcement, Panel B shows the result of the extended lockdown announcement following the initial announcement, and Panel C is our result of the main Shanghai lockdown announcement. Panel D and Panel E are the results of the lifting announcements, but Panel E is the result of the main Shanghai lockdown lifting announcement. <sup>a</sup>Obtained from Wilcoxon signed-rank tests. <sup>b</sup>Obtained from binomial sign tests. \* $p < 0.1$ , \*\* $p < 0.05$ , \*\*\* $p < 0.01$ .

| Event day(s)                                           | Mean   | $t$     | Median | $Z^a$  | % Negative | $Z^b$   |
|--------------------------------------------------------|--------|---------|--------|--------|------------|---------|
| Panel A: Lockdown announcement on March 27, 2022       |        |         |        |        |            |         |
| 0                                                      | 1.00%  | 0.87    | 1.09%  | 1.83*  | 0.00%      | -2.00** |
| 1                                                      | 0.04%  | 0.03    | -0.14% | 0.00   | 50.00%     | 0.00    |
| 2                                                      | -0.14% | -0.12   | -0.19% | -0.73  | 75.00%     | 1.00    |
| (-5, 0)                                                | 1.82%  | 0.65    | 0.03%  | 0.37   | 50.00%     | 0.00    |
| (0, 1)                                                 | 1.04%  | 0.64    | 0.43%  | 0.37   | 50.00%     | 0.00    |
| (0, 2)                                                 | 0.90%  | 0.45    | 0.51%  | 0.73   | 50.00%     | 0.00    |
| (0, 6)                                                 | -0.20% | -0.07   | 1.44%  | 0.37   | 25.00%     | -1.00   |
| Panel B: Lockdown announcement on March 31, 2022       |        |         |        |        |            |         |
| 0                                                      | -0.38% | -0.33   | -0.50% | -1.10  | 75.00%     | 1.00    |
| 1                                                      | -0.31% | -0.27   | 0.29%  | 0.00   | 50.00%     | 0.00    |
| 2                                                      | 0.79%  | 0.69    | -0.05% | 0.00   | 50.00%     | 0.00    |
| (-5, 0)                                                | -0.21% | -0.07   | -0.69% | 0.00   | 50.00%     | 0.00    |
| (0, 1)                                                 | -0.69% | -0.42   | 0.15%  | 0.37   | 25.00%     | -1.00   |
| (0, 2)                                                 | 0.11%  | 0.05    | 0.44%  | 0.00   | 50.00%     | 0.00    |
| (0, 6)                                                 | -2.54% | -0.84   | -1.24% | -1.83* | 100.00%    | 2.00**  |
| Panel C: Lockdown announcement on April 4, 2022 (main) |        |         |        |        |            |         |
| 0                                                      | 0.99%  | 0.86    | 0.34%  | 0.73   | 50.00%     | 0.00    |
| 1                                                      | -1.37% | -1.19   | -1.81% | -1.46  | 75.00%     | 1.00    |
| 2                                                      | -1.39% | -1.21   | -1.59% | -1.83* | 100.00%    | 2.00**  |
| (-5, 0)                                                | 1.17%  | 0.42    | 2.06%  | 0.73   | 25.00%     | -1.00   |
| (0, 1)                                                 | -0.38% | -0.24   | -0.29% | -0.73  | 50.00%     | 0.00    |
| (0, 2)                                                 | -1.78% | -0.89   | -1.71% | -0.73  | 50.00%     | 0.00    |
| (0, 6)                                                 | -1.75% | -0.58   | -0.29% | -0.37  | 50.00%     | 0.00    |
| Panel D: Lifting announcement on May 15, 2022          |        |         |        |        |            |         |
| 0                                                      | -2.34% | -2.04** | -2.00% | -1.83* | 100.00%    | 2.00**  |
| 1                                                      | -0.41% | -0.36   | -0.92% | -0.37  | 75.00%     | 1.00    |
| 2                                                      | -0.32% | -0.28   | -0.51% | -1.10  | 75.00%     | 1.00    |
| (-5, 0)                                                | -4.46% | -1.59   | -2.23% | -1.10  | 75.00%     | 1.00    |
| (0, 1)                                                 | -2.76% | -1.70*  | -2.24% | -1.83* | 100.00%    | 2.00**  |
| (0, 2)                                                 | -3.07% | -1.55   | -3.14% | -1.83* | 100.00%    | 2.00**  |
| (0, 6)                                                 | -5.70% | -1.88*  | -5.48% | -1.46  | 75.00%     | 1.00    |

Table B1 (Continued)

| Event day(s)                                         | Mean  | <i>t</i> | Median | <i>Z</i> <sup>a</sup> | % Negative | <i>Z</i> <sup>b</sup> |
|------------------------------------------------------|-------|----------|--------|-----------------------|------------|-----------------------|
| Panel E: Lifting announcement on May 30, 2022 (main) |       |          |        |                       |            |                       |
| 0                                                    | 0.46% | 0.40     | −0.02% | 0.00                  | 50.00%     | 0.00                  |
| 1                                                    | 0.48% | 0.42     | 0.01%  | 0.37                  | 50.00%     | 0.00                  |
| 2                                                    | 1.77% | 1.54     | 1.73%  | 1.46                  | 25.00%     | −1.00                 |
| (−5, 0)                                              | 3.21% | 1.14     | 5.23%  | 1.10                  | 25.00%     | −1.00                 |
| (0, 1)                                               | 0.94% | 0.58     | −0.31% | −0.37                 | 75.00%     | 1.00                  |
| (0, 2)                                               | 2.71% | 1.37     | 0.43%  | 0.73                  | 50.00%     | 0.00                  |
| (0, 6)                                               | 0.71% | 0.23     | −0.23% | 0.00                  | 50.00%     | 0.00                  |

Table B2 Aggregated abnormal returns of the suppliers of the four focal firms for all announcements (*N* = 252)

This table shows all the event study results for the five announcements. Panel A indicates the result of the initial but unclear Shanghai lockdown announcement, Panel B shows the result of the extended lockdown announcement following the initial announcement, and Panel C is our result of the main Shanghai lockdown announcement. Panel D and Panel E are the results of the lifting announcements, but Panel E is the result of the main Shanghai lockdown lifting announcement. <sup>a</sup>Obtained from Wilcoxon signed-rank tests. <sup>b</sup>Obtained from binomial sign tests. \**p* < 0.1, \*\**p* < 0.05, \*\*\**p* < 0.01.

| Event day(s)                                           | Mean   | <i>t</i> | Median | <i>Z</i> <sup>a</sup> | % Negative | <i>Z</i> <sup>b</sup> |
|--------------------------------------------------------|--------|----------|--------|-----------------------|------------|-----------------------|
| Panel A: Lockdown announcement on March 27, 2022       |        |          |        |                       |            |                       |
| 0                                                      | −0.43% | −1.08    | −0.25% | −3.78***              | 58.73%     | 2.77***               |
| 1                                                      | 0.41%  | 1.03     | 0.28%  | 2.56**                | 45.24%     | −1.51                 |
| 2                                                      | −0.48% | −1.19    | −0.25% | −3.91***              | 59.52%     | 3.02***               |
| (−5, 0)                                                | 0.00%  | 0.00     | 0.04%  | −0.63                 | 49.60%     | −0.13                 |
| (0, 1)                                                 | −0.02% | −0.03    | 0.12%  | −0.17                 | 48.02%     | −0.63                 |
| (0, 2)                                                 | −0.50% | −0.72    | −0.33% | −2.33**               | 55.56%     | 1.76*                 |
| (0, 6)                                                 | −2.01% | −1.89*   | −1.65% | −5.68***              | 65.48%     | 4.91***               |
| Panel B: Lockdown announcement on March 31, 2022       |        |          |        |                       |            |                       |
| 0                                                      | 0.23%  | 0.58     | 0.21%  | 2.24**                | 42.86%     | −2.27**               |
| 1                                                      | −0.52% | −1.29    | −0.36% | −4.41***              | 65.08%     | 4.79***               |
| 2                                                      | −0.29% | −0.72    | −0.26% | −3.86***              | 59.13%     | 2.90***               |
| (−5, 0)                                                | −0.34% | −0.35    | −0.23% | −1.23                 | 51.59%     | 0.50                  |
| (0, 1)                                                 | −0.29% | −0.51    | −0.18% | −1.35                 | 53.97%     | 1.26                  |
| (0, 2)                                                 | −0.57% | −0.83    | −0.65% | −3.29***              | 60.32%     | 3.28***               |
| (0, 6)                                                 | −2.46% | −2.32**  | −1.99% | −7.36***              | 72.22%     | 7.06***               |
| Panel C: Lockdown announcement on April 4, 2022 (main) |        |          |        |                       |            |                       |
| 0                                                      | −0.25% | −0.62    | −0.22% | −3.39***              | 58.73%     | 2.77***               |
| 1                                                      | −1.12% | −2.79*** | −0.92% | −9.02***              | 75.79%     | 8.19***               |
| 2                                                      | −0.29% | −0.72    | −0.14% | −2.48**               | 55.56%     | 1.76*                 |
| (−5, 0)                                                | −1.11% | −1.13    | −0.72% | −3.69***              | 60.71%     | 3.40***               |

**Table B2** (Continued)

| Event day(s)                                         | Mean   | <i>t</i> | Median | <i>Z</i> <sup>a</sup> | % Negative | <i>Z</i> <sup>b</sup> |
|------------------------------------------------------|--------|----------|--------|-----------------------|------------|-----------------------|
| (0, 1)                                               | −1.37% | −2.42**  | −1.35% | −7.80***              | 74.60%     | 7.81***               |
| (0, 2)                                               | −1.66% | −2.39**  | −1.52% | −7.32***              | 69.05%     | 6.05***               |
| (0, 6)                                               | −1.83% | −1.73*   | −1.45% | −5.39***              | 66.27%     | 5.17***               |
| Panel D: Lifting announcement on May 15, 2022        |        |          |        |                       |            |                       |
| 0                                                    | −0.51% | −1.27    | −0.40% | −4.25***              | 59.52%     | 3.02***               |
| 1                                                    | 0.53%  | 1.32     | 0.22%  | 3.62***               | 43.25%     | −2.14**               |
| 2                                                    | 0.75%  | 1.88*    | 0.46%  | 5.00***               | 36.11%     | −4.41***              |
| (−5, 0)                                              | 0.05%  | 0.05     | 0.58%  | 1.31                  | 45.63%     | −1.39                 |
| (0, 1)                                               | 0.01%  | 0.02     | −0.13% | −0.79                 | 51.98%     | 0.63                  |
| (0, 2)                                               | 0.76%  | 1.10     | 0.49%  | 2.86***               | 43.65%     | −2.02**               |
| (0, 6)                                               | 0.41%  | 0.39     | 0.01%  | 0.78                  | 49.21%     | −0.25                 |
| Panel E: Lifting announcement on May 30, 2022 (main) |        |          |        |                       |            |                       |
| 0                                                    | 0.71%  | 1.77*    | 0.57%  | 5.80***               | 34.13%     | −5.04***              |
| 1                                                    | 0.69%  | 1.72*    | 0.71%  | 5.15***               | 34.92%     | −4.79***              |
| 2                                                    | 0.67%  | 1.67*    | 0.45%  | 5.36***               | 35.32%     | −4.66***              |
| (−5, 0)                                              | 1.26%  | 1.28     | 0.76%  | 4.07***               | 39.68%     | −3.28***              |
| (0, 1)                                               | 1.40%  | 2.47**   | 1.29%  | 7.77***               | 28.17%     | −6.93***              |
| (0, 2)                                               | 2.05%  | 2.96***  | 1.94%  | 9.03***               | 23.81%     | −8.32***              |
| (0, 6)                                               | 3.20%  | 3.02***  | 2.52%  | 8.72***               | 23.41%     | −8.44***              |

**Table B3** Aggregated abnormal returns of suppliers contracting only four focal firms buyer companies for all announcements (*N* = 56)

This table shows all the event study results for the five announcements. Panel A indicates the result of the initial but unclear Shanghai lockdown announcement, Panel B shows the result of the extended lockdown announcement following the initial announcement, and Panel C is our result of the main Shanghai lockdown announcement. Panel D and Panel E are the results of the lifting announcements, but Panel E is the result of the main Shanghai lockdown lifting announcement. <sup>a</sup>Obtained from Wilcoxon signed-rank tests. <sup>b</sup>Obtained from binomial sign tests. \**p* < 0.1, \*\**p* < 0.05, \*\*\**p* < 0.01.

| Event day(s)                                     | Mean   | <i>t</i> | Median | <i>Z</i> <sup>a</sup> | % Negative | <i>Z</i> <sup>b</sup> |
|--------------------------------------------------|--------|----------|--------|-----------------------|------------|-----------------------|
| Panel A: Lockdown announcement on March 27, 2022 |        |          |        |                       |            |                       |
| 0                                                | −0.72% | −1.42    | −0.54% | −2.88***              | 66.07%     | 2.41**                |
| 1                                                | −0.22% | −0.43    | −0.40% | −1.31                 | 58.93%     | 1.34                  |
| 2                                                | −0.38% | −0.75    | −0.16% | −1.58                 | 55.36%     | 0.80                  |
| (−5, 0)                                          | −0.65% | −0.52    | −0.42% | −1.39                 | 53.57%     | 0.53                  |
| (0, 1)                                           | −0.93% | −1.30    | −1.05% | −3.05***              | 64.29%     | 2.14**                |
| (0, 2)                                           | −1.31% | −1.49    | −0.74% | −2.76***              | 60.71%     | 1.60                  |
| (0, 6)                                           | −3.98% | −2.97*** | −3.45% | −4.71***              | 76.79%     | 4.01***               |
| Panel B: Lockdown announcement on March 31, 2022 |        |          |        |                       |            |                       |
| 0                                                | −0.19% | −0.37    | −0.34% | −1.24                 | 62.50%     | 1.87*                 |
| 1                                                | −0.80% | −1.57    | −0.88% | −3.17***              | 75.00%     | 3.74***               |

Table B3 (Continued)

| Event day(s)                                           | Mean   | <i>t</i> | Median | <i>Z</i> <sup>a</sup> | % Negative | <i>Z</i> <sup>b</sup> |
|--------------------------------------------------------|--------|----------|--------|-----------------------|------------|-----------------------|
| 2                                                      | −0.54% | −1.06    | −0.59% | −2.48**               | 62.50%     | 1.87**                |
| (−5, 0)                                                | −2.49% | −2.01**  | −2.58% | −4.35***              | 78.57%     | 4.28***               |
| (0, 1)                                                 | −0.98% | −1.37    | −1.08% | −2.73***              | 67.86%     | 2.67***               |
| (0, 2)                                                 | −1.52% | −1.73*   | −1.50% | −3.32***              | 76.79%     | 4.01***               |
| (0, 6)                                                 | −3.86% | −2.88*** | −3.20% | −4.56***              | 78.57%     | 4.28***               |
| Panel C: Lockdown announcement on April 4, 2022 (main) |        |          |        |                       |            |                       |
| 0                                                      | −0.50% | −0.99    | −0.48% | −2.07**               | 62.50%     | 1.87*                 |
| 1                                                      | −1.21% | −2.39**  | −0.88% | −4.83***              | 75.00%     | 3.74***               |
| 2                                                      | −0.61% | −1.21    | −0.23% | −2.15**               | 57.14%     | 1.07                  |
| (−5, 0)                                                | −2.82% | −1.68*   | −2.67% | −3.87***              | 75.00%     | 3.74***               |
| (0, 1)                                                 | −1.71% | −2.39**  | −1.31% | −4.04***              | 75.00%     | 3.74***               |
| (0, 2)                                                 | −2.33% | −2.65*** | −1.38% | −4.63***              | 78.57%     | 4.28***               |
| (0, 6)                                                 | −2.71% | −2.02**  | −1.27% | −3.00***              | 66.07%     | 2.41**                |
| Panel D: Lifting announcement on May 15, 2022          |        |          |        |                       |            |                       |
| 0                                                      | −0.31% | −0.61    | 0.10%  | −0.07                 | 42.86%     | −1.07                 |
| 1                                                      | 0.21%  | 0.41     | 0.09%  | 1.04                  | 48.21%     | −0.27                 |
| 2                                                      | 0.95%  | 1.87**   | 0.53%  | 2.50**                | 37.50%     | −1.87*                |
| (−5, 0)                                                | −0.76% | −0.61    | 0.02%  | −0.14                 | 50.00%     | 0.00                  |
| (0, 1)                                                 | −0.10% | −0.14    | 0.40%  | 0.33                  | 42.86%     | −1.07                 |
| (0, 2)                                                 | 0.85%  | 0.97     | 0.98%  | 2.10**                | 41.07%     | −1.34                 |
| (0, 6)                                                 | 0.46%  | 0.34     | 0.13%  | 0.30                  | 48.21%     | −0.27                 |
| Panel E: Lifting announcement on May 30, 2022 (main)   |        |          |        |                       |            |                       |
| 0                                                      | 0.29%  | 0.57     | 0.57%  | 1.56                  | 42.86%     | −1.07                 |
| 1                                                      | −0.10% | −0.20    | 0.10%  | −0.01                 | 48.21%     | −0.27                 |
| 2                                                      | 0.90%  | 1.78*    | 0.51%  | 2.64***               | 37.50%     | −1.87*                |
| (−5, 0)                                                | 0.67%  | 0.54     | 0.25%  | 0.38                  | 46.43%     | −0.53                 |
| (0, 1)                                                 | 0.19%  | 0.26     | 0.53%  | 1.32                  | 37.50%     | −1.87*                |
| (0, 2)                                                 | 1.09%  | 1.24     | 1.55%  | 3.16***               | 32.14%     | −2.67***              |
| (0, 6)                                                 | 2.39%  | 1.78*    | 1.54%  | 2.47**                | 33.93%     | −2.41**               |